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Statistics: is the process of collecting, organizing, analyzing and interpreting data so that we can understand the data and make decisions based on it

Types of Statistics: statistics is mainly divided into Descriptive Statistics and Inferential Statistics

Descriptive Statistics: is used to describe and summarize the data we already have using measures like mean, median, mode, range, variance, standard deviation and also charts/graphs. i.e if we have marks of 100 students, finding the average marks of those 100 students is descriptive statistics

Inferential Statistics: is used to take a sample from a larger population and use the sample results to make conclusions/predictions about the entire population. i.e instead of asking 10,000 customers about a product, we take 500 customers as a sample and use their responses to understand the overall customer opinion

Population: is the complete group/data that we are interested in studying i.e all students in a university

Sample: is a smaller part selected from the population for analysis i.e selecting 200 students from the university

Data: is the information/values collected for analysis and it can mainly be categorical or numerical

Categorical Data/Qualitative Data: represents categories, names or labels rather than measurable numerical quantities i.e gender, city, department, product category. Categorical data is mainly divided into nominal and ordinal

Nominal Data: is categorical data where there is no meaningful order/ranking between the categories i.e blood group A/B/AB/O, colors red/blue/green, city Dallas/Houston/Austin

Ordinal Data: is categorical data where categories have a meaningful order/ranking but the exact difference between the levels cannot be measured i.e customer satisfaction Poor/Average/Good/Excellent or education level High School/Bachelors/Masters

Numerical Data/Quantitative Data: represents measurable/countable numerical values on which mathematical calculations make sense i.e age, salary, height, number of customers. Numerical data can be discrete or continuous and its measurement levels include interval and ratio

Interval Data: is numerical data where the difference between values is meaningful but there is no true/absolute zero. i.e temperature in Celsius where 0°C does not mean there is no temperature. Because there is no true zero, saying 20°C is twice as hot as 10°C is not meaningful

Ratio Data: is numerical data where differences are meaningful and there is a true zero which represents absence of the quantity i.e salary, weight, height, distance, age. If weight is 0 it means no weight, and 100 kg is meaningfully twice 50 kg

Measures of Central Tendency: are used to find the central/typical value of the data. Main measures are mean, median and mode

Mean: is the average of all values. It is calculated by adding all the values and dividing by total number of values i.e 10,20,30 → mean=(10+20+30)/3=20

Median: is the middle value after arranging the data in ascending/descending order. i.e 10,20,30 → median=20. If there are even number of values, median is the average of the two middle values i.e 10,20,30,40 → median=(20+30)/2=25

Mode: is the value which occurs most frequently in the data i.e 10,20,20,30 → mode=20

Measures of Dispersion: are used to understand how spread out/variable the data is around the central value. Main measures include range, variance and standard deviation

Range: is the difference between maximum and minimum value i.e 10,20,30,50 → range=50-10=40

Variance: measures how far the values are spread from the mean. Higher variance means values are more spread out and lower variance means values are closer to the mean

Standard Deviation: measures how much the values typically differ from the mean and is the square root of variance. Small standard deviation means values are closer to the mean whereas large standard deviation means values are more spread out

Random Variable: is a variable whose numerical value depends on the outcome of a random event/experiment. Random variables are mainly discrete or continuous

Discrete Random Variable: can take separate/countable values, usually whole-number counts i.e number of customers entering a store can be 10,11,12 but cannot be 10.5 customers

Continuous Random Variable: can take any value within a particular range including decimal values i.e height can be 5.7 feet, weight can be 72.5 kg, temperature can be 98.6°F

Discrete vs Continuous: discrete data is countable and normally has separate values whereas continuous data is measurable and can have any value within a range

Skewness: tells us about the asymmetry/shape of the distribution and whether the data is stretched more toward the left or right side

Zero Skewness/Symmetric Distribution: data is approximately equally distributed on both sides with no long tail toward either side. In a perfectly symmetric distribution mean=median=mode

Right Skewed/Positive Skewness: distribution has a longer tail toward the right/higher values. A few very high values pull the mean toward the right, so generally Mode < Median < Mean. Salary/income data is a common example because a small number of people can have extremely high incomes

Left Skewed/Negative Skewness: distribution has a longer tail toward the left/lower values. A few very low values pull the mean toward the left, so generally Mean < Median < Mode

Important point for Skewness: the direction of skewness is identified by the direction of the long tail, not by where most of the data is located. Right tail = positive/right skew and left tail = negative/left skew

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Z-score: is used to understand how far a particular data value is from the mean in terms of standard deviations. Z-score=0 means the value is equal to the mean, positive Z-score means the value is above the mean and negative Z-score means the value is below the mean

Z-score formula: Z=(x-mean)/standard deviation where x is the individual value, mean is the average and standard deviation tells how spread out the data is

Example of Z-score: if mean=50, standard deviation=10 and x=70 then Z=(70-50)/10=2 which means 70 is 2 standard deviations above the mean

Outliers: are data points which are unusually far away from most of the other values in the dataset i.e if salaries are 40000,45000,50000,55000,500000 then 500000 could be an outlier

Z-score for detecting outliers: one common rule is values having Z-score greater than 3 or less than -3 can be considered potential outliers, but this is only a guideline and depends on the dataset/domain

Handle outliers in the dataset: before handling an outlier we should understand why it exists because an outlier may be an error or it may be a genuine extreme value. Common methods are removing, capping/winsorizing, transformations, imputation and binning

Remove outliers: we can remove the rows containing outliers when they are clearly incorrect/unwanted and removing them does not cause important information loss

Capping/Winsorizing: instead of deleting outliers we limit extreme values to selected lower/upper boundaries. Values above the upper limit are replaced with the upper limit and values below the lower limit are replaced with the lower limit

Log Transformation: is used to reduce the effect of very large values and right-skewed data by applying a logarithm to the values. It can make highly skewed data more manageable, but it is not suitable for every dataset and needs care with zero/negative values

Imputation: is the process of replacing missing values, and in some workflows extreme values may first be treated as invalid/missing and then imputed. Mean, median or mode can be used depending on the data

Mean Imputation: missing values can be replaced with the mean of the column. It is generally more suitable when numerical data is reasonably symmetric and does not contain strong outliers

Median Imputation: missing values can be replaced with the median. Median is less affected by extreme values so it is often preferred for skewed numerical data

Mode Imputation: missing values can be replaced with the most frequently occurring value and is commonly useful for categorical/discrete data

Binning: is the process of grouping continuous numerical values into intervals/categories i.e age 0-18=Young, 19-40=Adult, 41-60=Middle Age, 60+=Senior. It can simplify data and reduce the effect of small variations/noise

Probability: tells us how likely an event is to happen. Probability ranges from 0 to 1 where 0 means impossible and 1 means certain

Probability Distribution: describes how probabilities are distributed across the possible values/outcomes of a random variable

Normal Distribution: is a continuous probability distribution which has a symmetric bell-shaped curve where most values are around the mean and fewer values occur farther from the mean. In an ideal normal distribution mean=median=mode

Bernoulli Distribution: represents a single experiment/trial which has only two possible outcomes generally success/failure represented as 1 and 0 i.e tossing a coin once and checking whether it is Heads or not

Binomial Distribution: represents the number of successes in a fixed number of independent Bernoulli trials where each trial has the same probability of success i.e tossing a coin 10 times and counting how many Heads occur

Bernoulli vs Binomial: Bernoulli deals with one trial having two outcomes whereas Binomial deals with the number of successes across multiple independent Bernoulli trials

Uniform Distribution: is a distribution where all values/outcomes within the defined range have equal or constant probability density. A fair die is a simple discrete uniform example because each number from 1 to 6 has the same probability

Poisson Distribution: is a discrete probability distribution used to model the number of events occurring within a fixed interval of time, space or another exposure when events occur with a

stable average rate under the model assumptions i.e number of calls received by a call center in one hour

Central Limit Theorem(CLT): states that under common conditions, when we repeatedly take sufficiently large independent samples from a population and calculate their sample means, the distribution of those sample means becomes approximately normal even when the original population is not normally distributed

Inferential Statistics: is used to draw conclusions or make predictions about a population using information collected from a sample. It commonly includes estimation, confidence intervals and hypothesis testing

Sampling: is the process of selecting a smaller group/sample from a larger population so that we can study the sample and make conclusions about the population

Sampling Techniques: are different methods used to select a sample from the population. They can broadly be divided into probability and non-probability sampling

Probability Sampling/Random Sampling: every member of the population has a known non-zero chance of being selected through a random mechanism, which helps reduce selection bias

Simple Random Sampling: every individual has an equal chance of being selected from the population i.e randomly selecting 100 employee IDs from 1000 employees

Systematic Random Sampling: we select members using a fixed interval after choosing a random starting point i.e from 1000 customers select a random starting customer and then select every 10th customer

Stratified Sampling: population is divided into meaningful groups called strata based on characteristics such as department, age group or region and then random samples are selected from each group. This helps ensure important groups are represented in the sample

Hypothesis Testing: is a statistical method used to evaluate evidence from sample data and decide whether there is enough evidence to reject a claim/assumption about the population

Null Hypothesis(H0): is the default assumption/claim that there is no effect, no difference or no relationship i.e a new training method does not change the average employee performance

Alternative Hypothesis(H1/Ha): is the competing claim that there is an effect, difference or relationship i.e the new training method changes the average employee performance

Type 1 Error: happens when we reject a null hypothesis even though the null hypothesis is actually true. In simple terms we conclude that an effect/difference exists when it actually does not exist. It is also called a false positive

Type 2 Error: happens when we fail to reject the null hypothesis even though the null hypothesis is actually false. In simple terms an effect/difference actually exists but our test fails to detect it. It is also called a false negative

Type 1 vs Type 2: Type 1=reject a true null hypothesis/false positive whereas Type 2=fail to reject a false null hypothesis/false negative